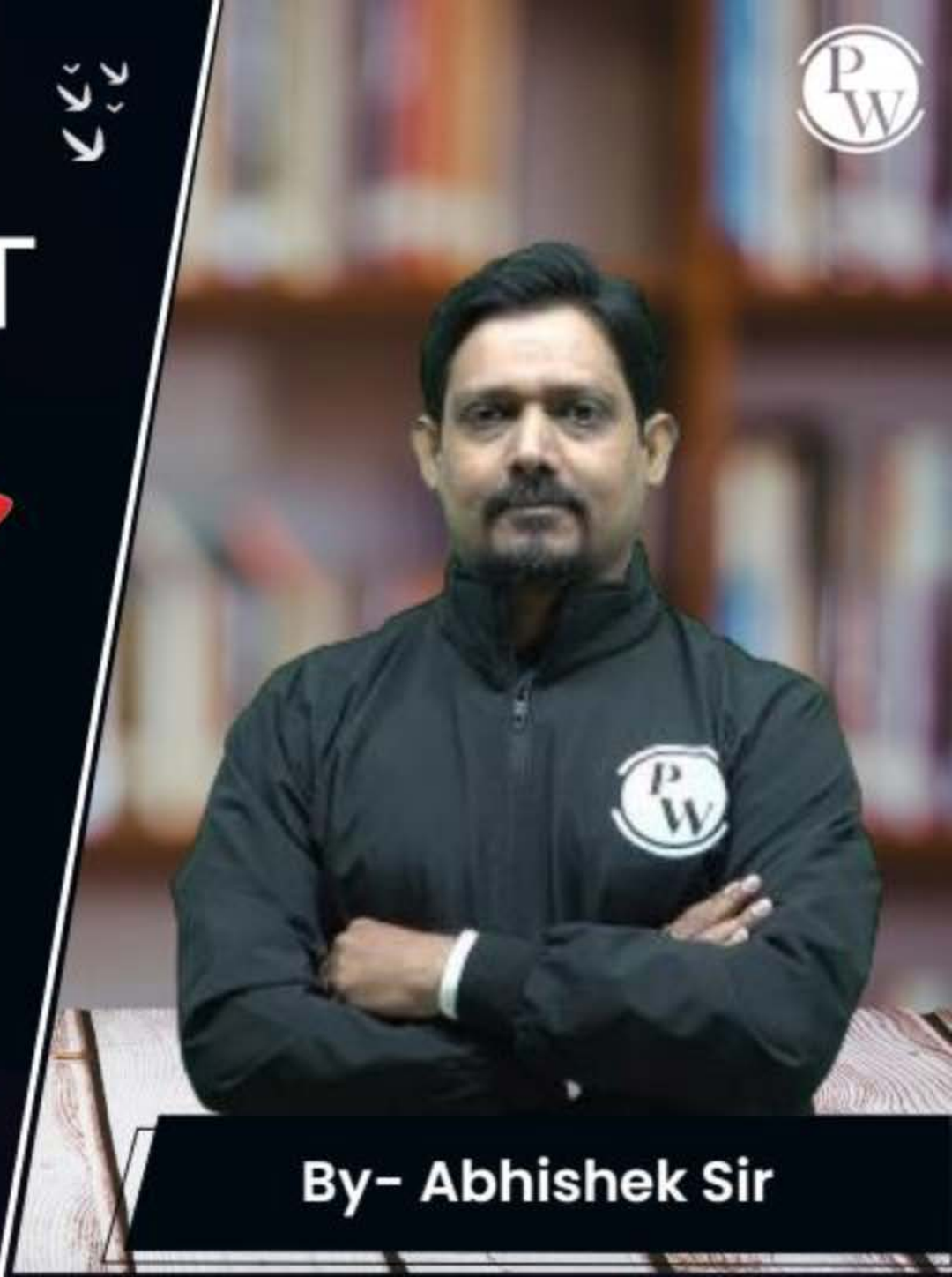
Computer Science & IT



Fundamental of C Language

Array & Pointers

Lecture No. 01



By- Abhishek Sir

Recap of Previous Lecture



- Topic — Storage class
- Topic — auto
- Topic — static
- Topic — Register
- Topic — Extern

auto, static, Register, Extern } Memory Layout + Local & global

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

pointer

Call by reference

Slide



Pointers

Embedded System

C - middle Level Language :- C is having high level programming construct

C is also low level program construct



Pointers

C - Laguage allows a variable to access address of it

int a = 10;



100 ← Addres of Memory Location

Assumed value

Slide



Pointers

Pointer variable : pointer variable is a variable that holds address of another variable of same data type



Pointers

Declaration of panter variable :-

* is a dereference operator

```
int a;
int *ptr;
```

← Declaration of pointer variable

Slide



Pointers

Assignment of panter variable :

```
int a = 10;
int * ptr;
ptr = & a;
```



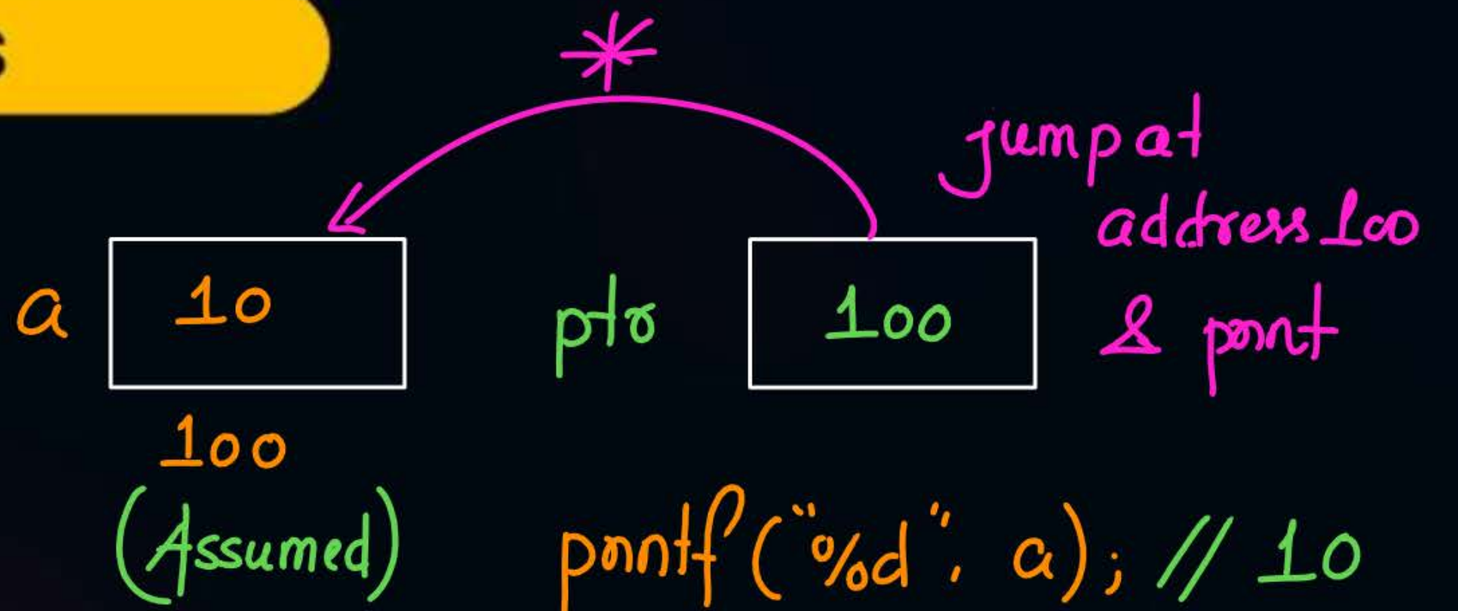
&a is having value 100

& is Address of operator appiled to variable and it access address of a variable

Slide

Pointers

```
int a = 10,
int *ptr,
ptr = &a.
```




```
printf("%d", a); // 10
printf("%u", &a); // Address of a, 100
printf("%u", ptr), // 100
printf("%d", *ptr), // 10
```

Address/ pointer

format specifier

%p Hexadecimal

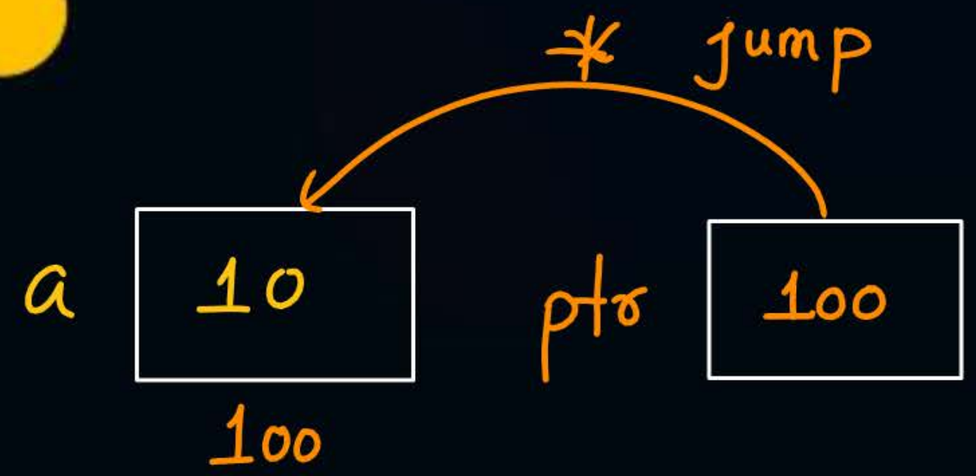
%u is also used to get unsigned

Slide



Pointers

```
#include<stdio.h>
int main() {
    int a=10,
    int *ptr;
    ptr = &a;
    a = a+20,
    *ptr = *ptr+10,
    printf("%d", a),
    return 0;
}
```

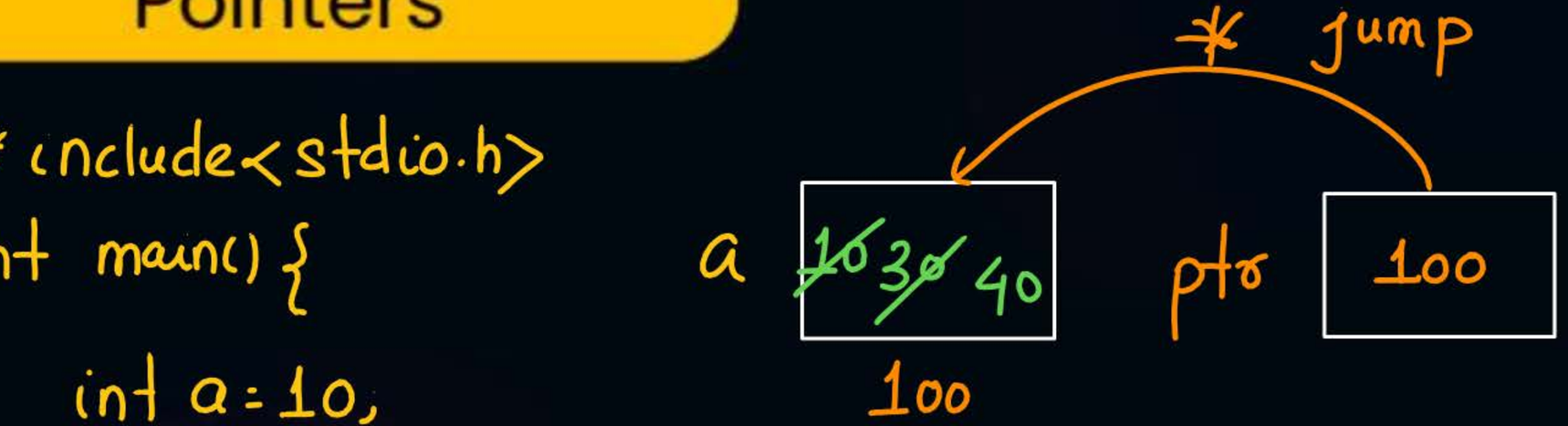


whenever a pointer variable contains address of another variable then a relationship build between two variables.

We can use both variable to modify same Memory Location

Slide

Pointers




```
#include<stdio.h>
int main(){
    int a=10;
    int *ptr;
    ptr = &a;
    a=a+20;
    *ptr=*ptr+10;  // 30+10=40
    printf("%d", a); //(40)
    return 0;
}
```

Slide



Pointers

```
#include<stdio.h>
                a=10, b=20
void swap(int a, int b){
    int temp;
    temp = a; // temp=10
    a = b,  // a=20
    b = temp; // b=10
}
```

a 20 b 10

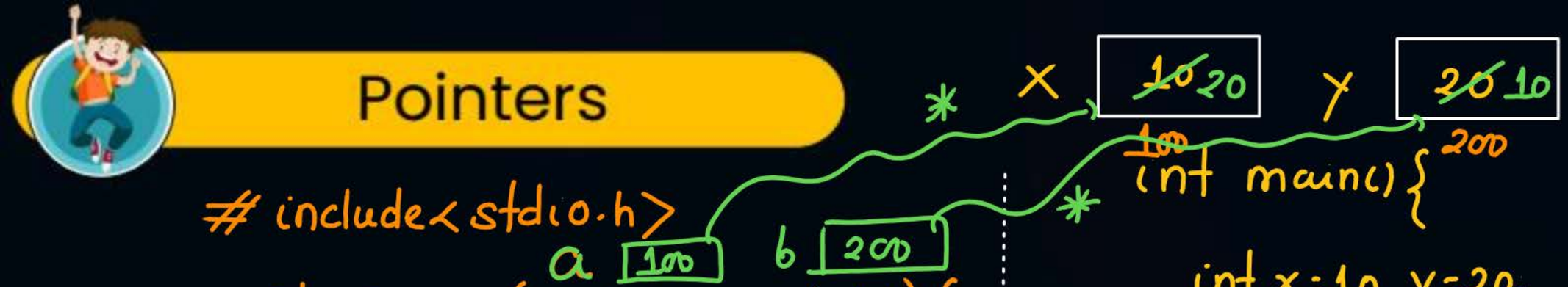
Local Swap

```
int main() {
    int x=10, y=20,
        10,20
    swap(x,y);
    printf("%d", x), // 10
    printf("%d", y). // 20
    return 0;
}
```

x 10 y 20

Slide

Pointers




```
#include<stdio.h>
void swap( int *a, int*b){

    int temp;

    temp = *a; // temp: 10

    *a = *b;

    *b = temp

}
```

```
int main(){

int x=10, y=20.

Swap(&x, &y).

printf("%d", x); // 20

printf("%d", y); // 10

return 0,

}
```

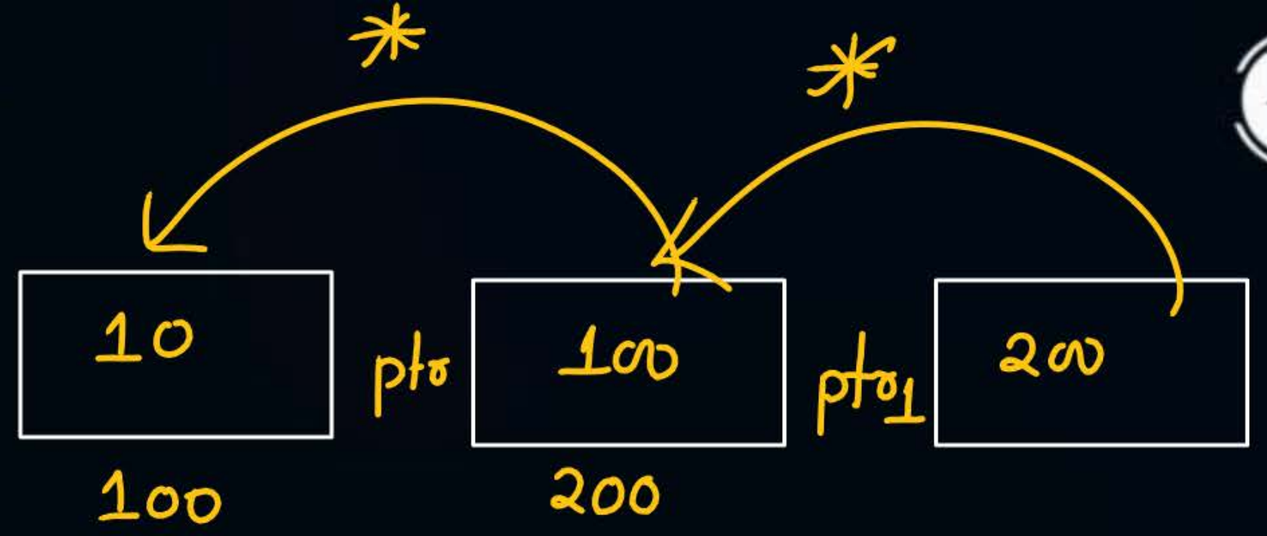
Call by reference

Slide



Pointers to Pointer

```
int a = 10
int *ptr;
int **ptr1;
ptr = &a;
ptr1 = &ptr;
```



Slide



2 mins Summary

Topic — pointer

Topic — Swap

Topic — Call by reference

Topic — Double pointer

Topic

Slide

THANK - YOU

